

SE212 Database System and Design

Lab 10: Answering business questions with SQL (20 points)

You have just joined the Accounts Payable team of a small publishing company as a data analyst. The company's data is in the **ap_extra** database you installed in Lab 8. Your manager sends you the questions below.

For each question, **you** find out which tables and columns hold the answer, how they connect, and write the SQL. Write all your answers in one SQL file, **Lab10_<SID>.sql**: the query under the question number, and anything the question asks you to explain as a comment.

Before the lab: explore the database

1. Start MySQL Server and open MySQL Workbench (Lab 8, Option A or B).
2. **USE ap_extra;** then explore it: **SHOW TABLES;;**, **DESCRIBE <table>;**, and **SELECT * FROM <table> LIMIT 10;** on each table.
3. Open the ER diagram **ER-diagram-APdatabaseW2ExtraTables-Murach.pdf** in the **Lab8_murach** folder. Work out the primary and foreign keys before you start.
4. If **ap_extra** is missing, run **create_databases.sql** from **Lab8_murach** again, as in Lab 8 Part 2.

Every result must have column names your manager can read (use **AS**).

Part 1 (4 points)

1.1 (1 point) Which invoices are not fully paid yet? For each, show the vendor, the invoice number, the invoice date and how much we still owe. Biggest amount first.

1.2 (1 point)

- a) The auditor wants every invoice from a vendor with **“Inc”** anywhere in its name: vendor, invoice number and amount, sorted by vendor.
- b) Zylka Design numbers its invoices **97/** followed by **exactly three characters**. List them. One Zylka invoice starting with **97/** does **not** follow that rule. Which one? Answer in a comment.

1.3 (2 points) Marketing wants every invoice line charged to an **advertising** account: the vendor, the invoice number, what the line was for, the account name and the amount. Biggest first. Take a screenshot (**S1**).

Part 2 (5 points)

2.1 (2 points) For **every** vendor in **Ohio**, show the city and how many invoices they have sent us. Vendors that never sent one must show **0**. Most active first.

2.2 (2 points) Accounting wants to clean up the chart of accounts. Which accounts with **“Tax”** in their description have **never** been used on any invoice line? How many accounts in total (tax or not) have never been used? One account with **“Tax”** in its description is not in your list: which one, and why? Answer both in comments.

2.3 (1 point) Customer service wants **one list** with **every** person in the old vendor contact list, even if their vendor never invoiced us, **and every** invoice, even if we have no contact person for that vendor, matched up where possible. Show the vendor id and last name of the contact, and the invoice id. How many rows are in the list? Answer in a comment. (*MySQL does not support FULL OUTER JOIN.*)

Part 3 (3 points)

3.1 (1 point) The name of each vendor's contact person is stored in two different places. Write a query with a **NATURAL JOIN** that shows both names side by side with the vendor name. For how many vendors do the two names disagree, and which column did the **NATURAL JOIN** use? Answer in a comment.

3.2 (1 point) Purchasing wants a planning sheet with **every** contact person paired with **every** payment term we offer. How many rows, and why exactly that number? Answer in a comment.

3.3 (1 point) A colleague wanted to count vendors with their default payment term and wrote:

```
SELECT COUNT(*) FROM vendors NATURAL JOIN terms;
```

Write the correct query. In a comment: what number does the colleague's query give, what is the correct number, and why is the colleague's query wrong?

Part 4 (8 points)

4.1 (2 points) Who are our biggest suppliers? For every vendor that sent us **more than one** invoice, show how many invoices, the total amount invoiced and how much we still owe them. Biggest total first. Take a screenshot (**S2**).

4.2 (2 points) Where does our money go? For each account, show how many invoice lines were charged to it and the total amount. Only accounts with **more than 5,000** in total, biggest first.

4.3 (2 points) Which vendors do we pay late? For every vendor we have paid **after the due date** at least once, show how many of its invoices we paid late and, on average, how many days late. Most late invoices first. How many invoices did we pay late in total? Answer in a comment.

4.4 (2 points) Your own question. Think of **one business question** a manager at this company might ask, and answer it with **one** query. It must use **at least two tables** and **GROUP BY**. Write the question as a comment above the query, and one sentence below it saying what the answer means for the business.

Submission checklist

Submit on Mango (CMU Canvas), assignment **Lab 10**, before the deadline. Individual work. Upload two files named with your student ID:

File 1: Lab10_<SID>.sql, one SQL script that runs from top to bottom **without any error**. Put a comment header before each answer, and write anything a question asks you to explain as a comment. If you try a query that gives an error (for example **FULL OUTER JOIN**), comment it out so the script still runs:

```
-- Lab 10 | Name: <your name> | SID: <your SID>
USE ap_extra;
```

```
-- ===== 1.1 =====
-- SELECT here
-- your explanation, if the question asks for one
```

```
-- ===== 1.2 =====
```

```
...
```

```
-- ===== 4.4 =====
```

```
-- Question: <your business question>
-- SELECT here
-- Answer: <what it means for the business>
```

File 2: Lab10_<SID>_screenshots.pdf (or S1.png, S2.png): the whole Workbench window with the query and its result grid, for **S1** (1.3) and **S2** (4.1).

Marking. Each task is marked on the result (right rows, readable column names) and on your answers. A script that stops with an error loses the points for every task after the error, so run the whole file once before you submit.